

Lecture 3: Affective Computing

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

- 1- Emotional Intelligence (affective computing)

- Session Learning Outcomes

By the end of today's lecture you will be able to:

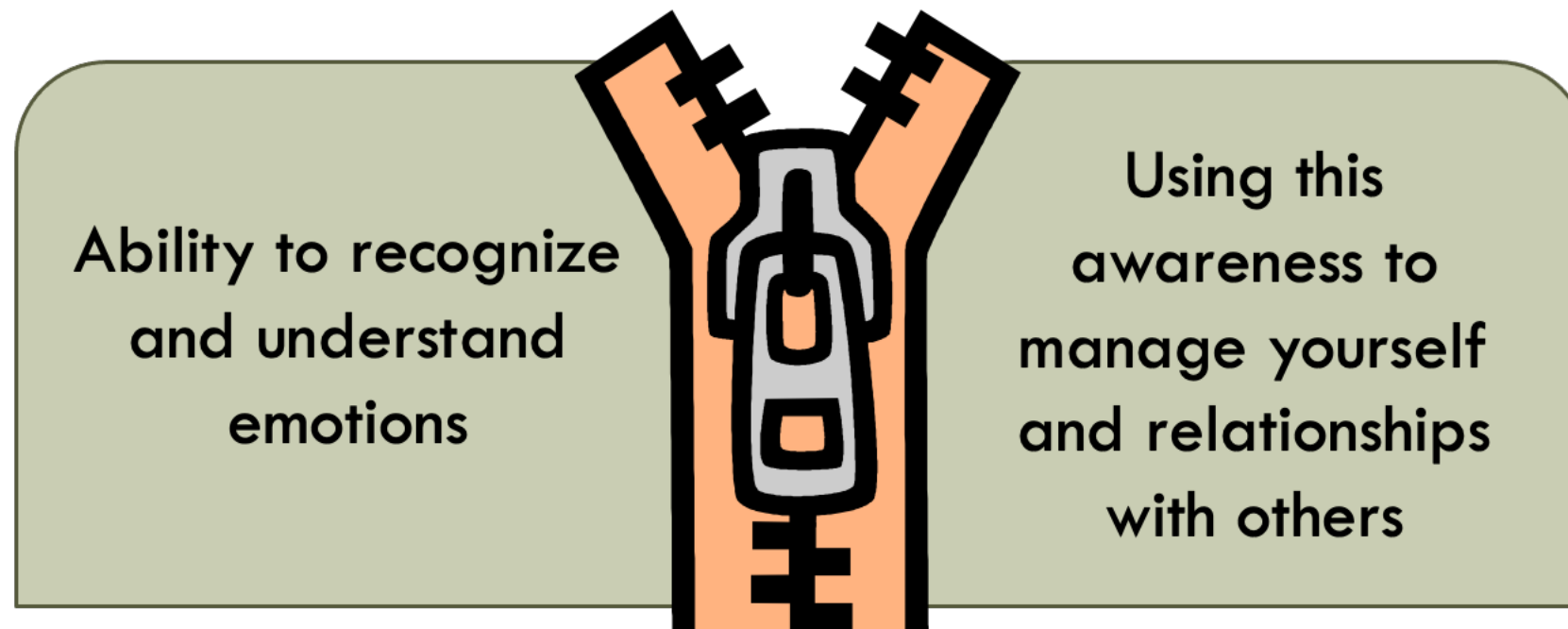
- 1- Describe affective computing

- 2- Describe the cues used for emotion recognition in speech

- 3- Discuss emotion recognition through facial expressions

Affective Computing

- Affective computing is a subset of artificial intelligence that **measures, understands, simulates, and reacts** to human emotions. It's also known as **artificial emotional intelligence**.
- Affective computing is the study and development of systems and devices that can **recognize, interpret, process, and simulate** human affects. It is an interdisciplinary field spanning computer science, psychology, and cognitive science.



Emotional Intelligence in Humans

Affective Computing



Weak Management of Emotional State

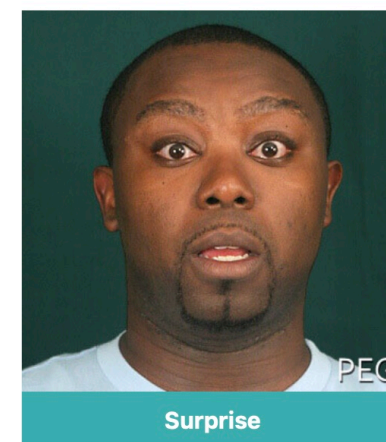
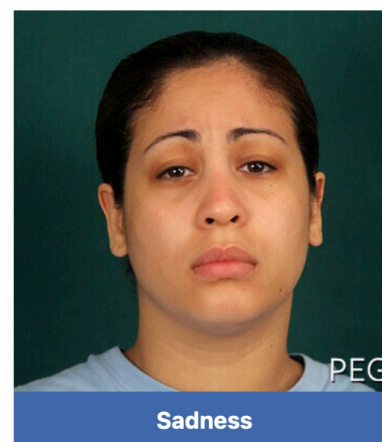
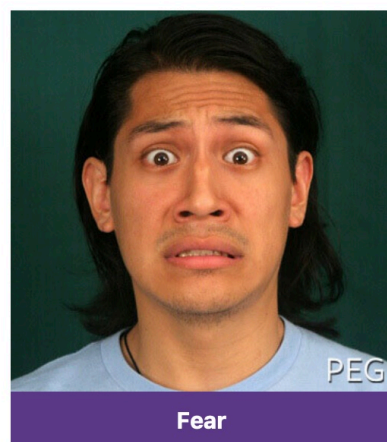
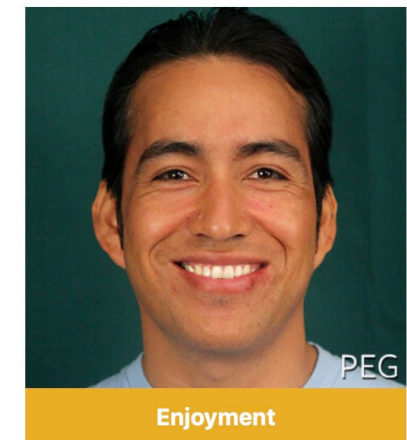
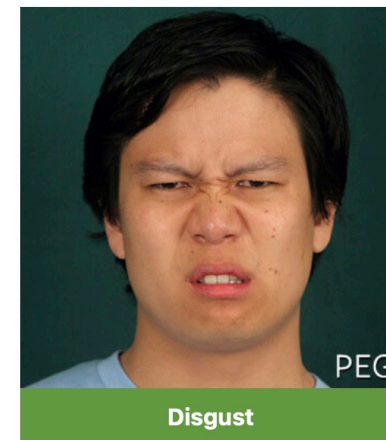
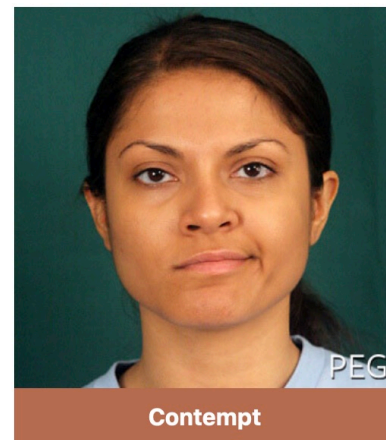
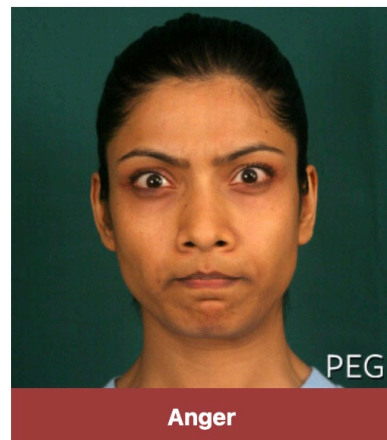
Affective Computing

- More about Emotions

1. Emotions are mental states brought on by neurophysiological changes, variously associated with **thoughts, feelings, behavioral responses**, and a degree of pleasure or displeasure.

2. Several models of emotion exist, but some basic ones are:

I. Ekman: identified seven basic emotions: **anger, surprise, disgust, enjoyment, fear, sadness, and contempt.**



Affective Computing

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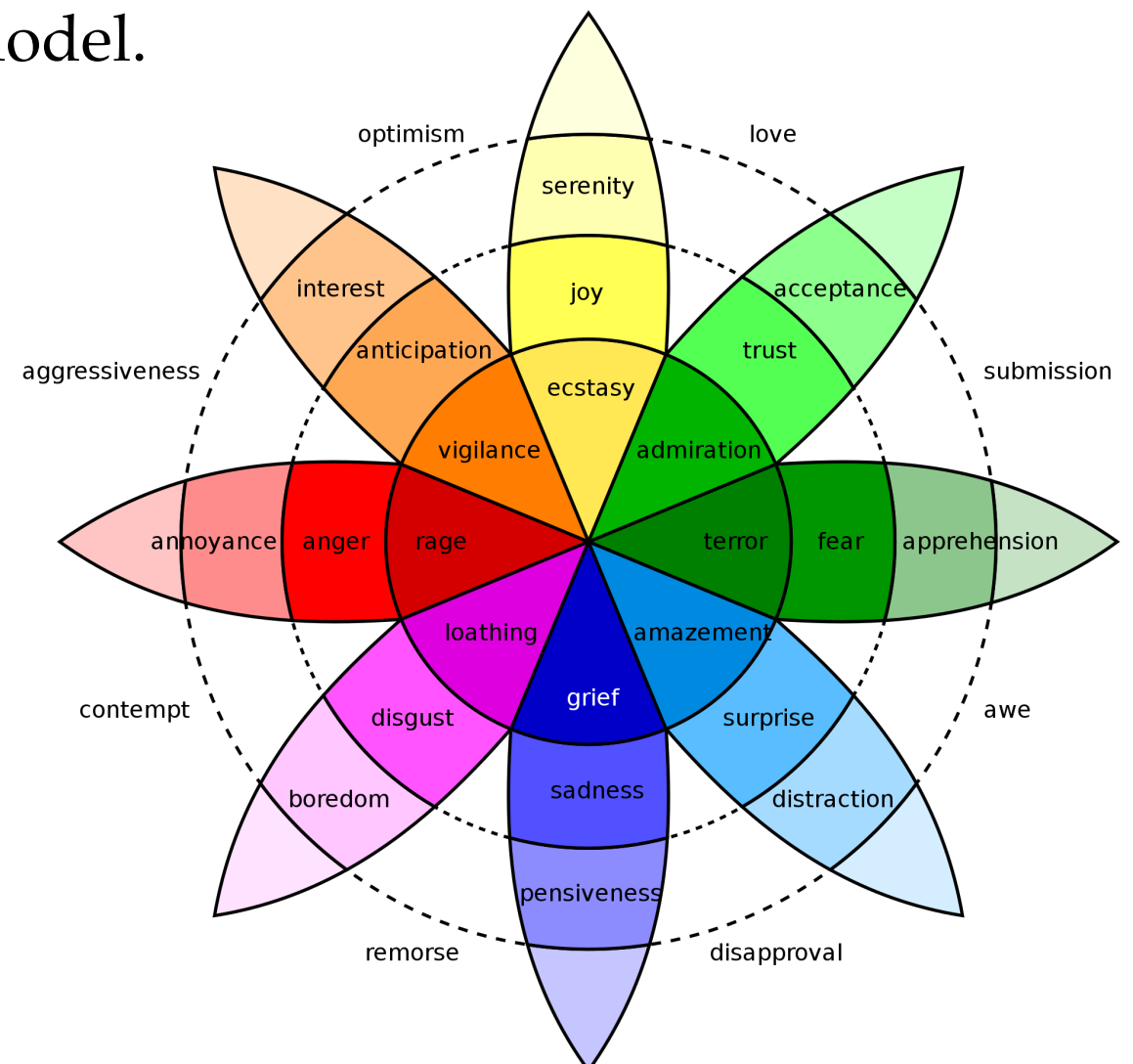
II. Plutchik: identified eight basic emotions: **acceptance, fear, surprise, sadness, disgust, anger, anticipation**, and **joy**, besides a group of secondary emotions in a wheel model.

- Primary combined emotions

Joy + Trust = Love

- Secondary combined emotions

Joy + Anger = Pride

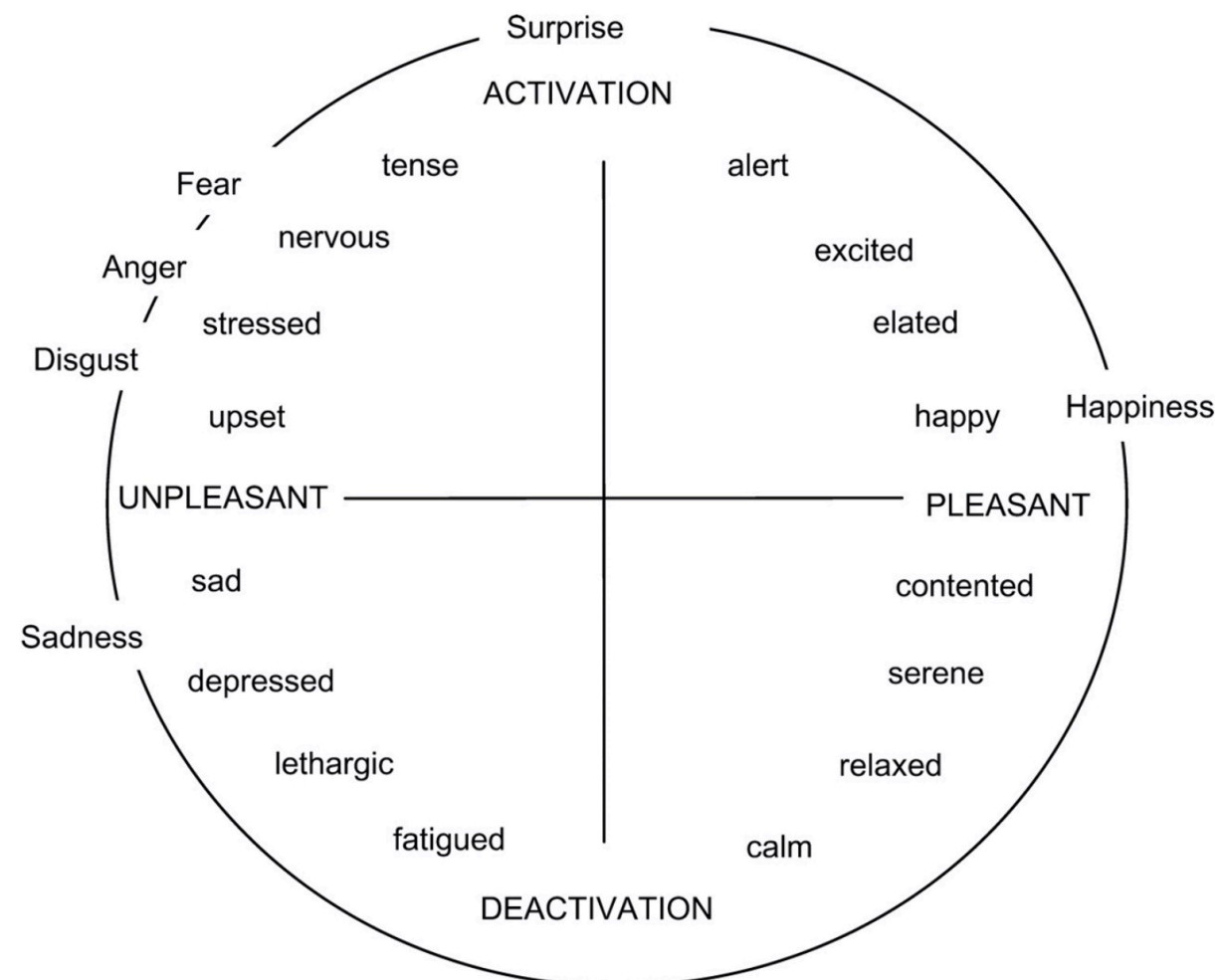


Affective Computing

- More about Emotions

1. Emotions are mental states brought on by neurophysiological changes, variously associated with **thoughts, feelings, behavioral responses**, and a degree of pleasure or displeasure.
2. Several models of emotion exist, but some basic ones are:

III. Russell: suggested the circumplex model (bipolar dimensions), where all emotions are arranged in a circle controlled by two independent dimensions: **hedonic** (pleasure-displeasure) and **arousal** (activation)



Affective Computing

Which one is more intelligent?



Even a puppy can tell when you are angry with it.

(Nicholas Negroponte, *Being Digital*)

Affective Computing

Computer will keep showing you the same data, whether you look like this, or like this



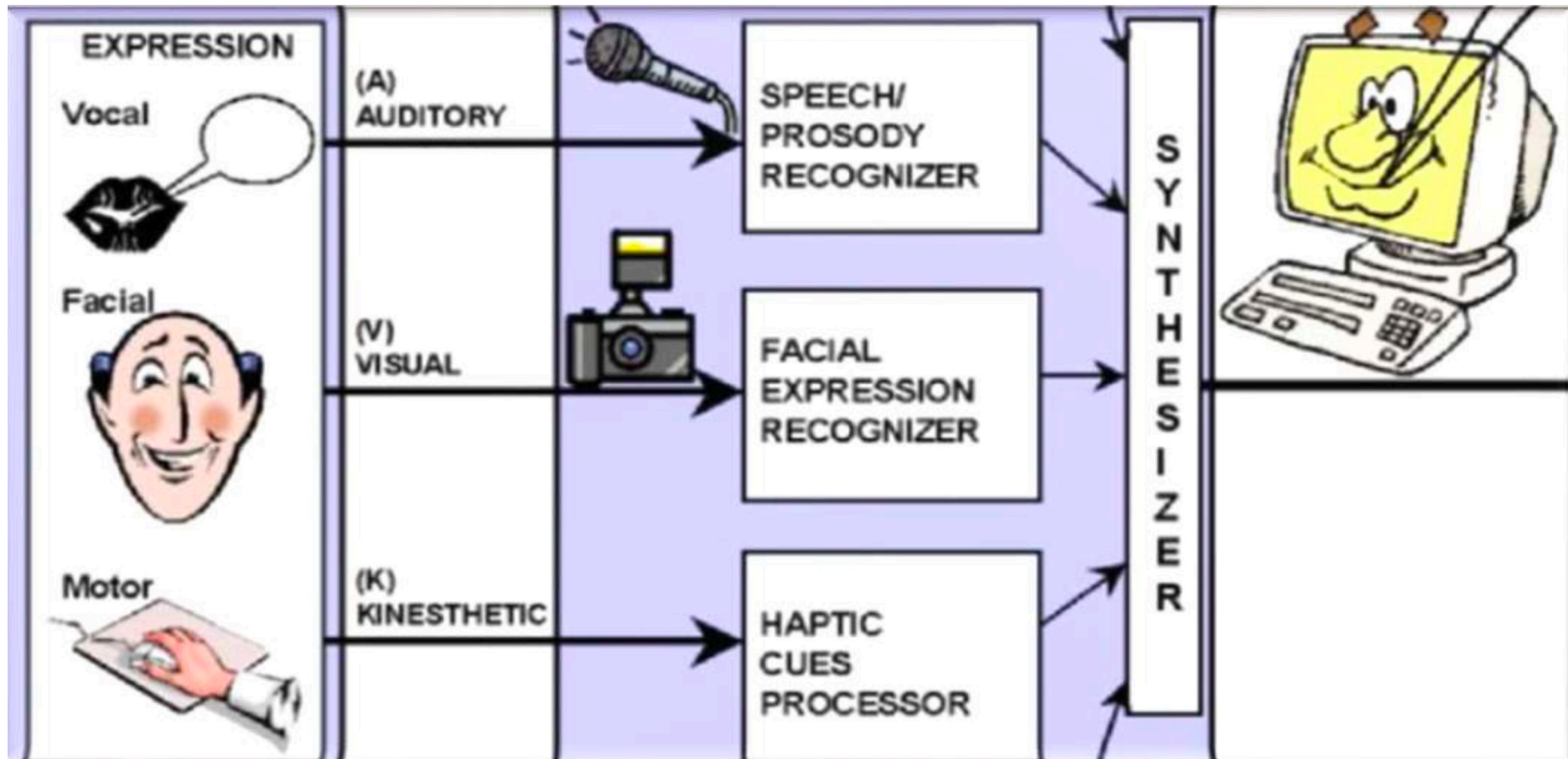
Affective Computing

“The question is not whether intelligent machines can have any emotions, but whether machines can be intelligent without any emotions.”

- — Marvin Minsky (1927–2016) (Co-founder of AI Lab at MIT, Turing Award winner (most prestigious award in Computer Science)).

Affective Computing

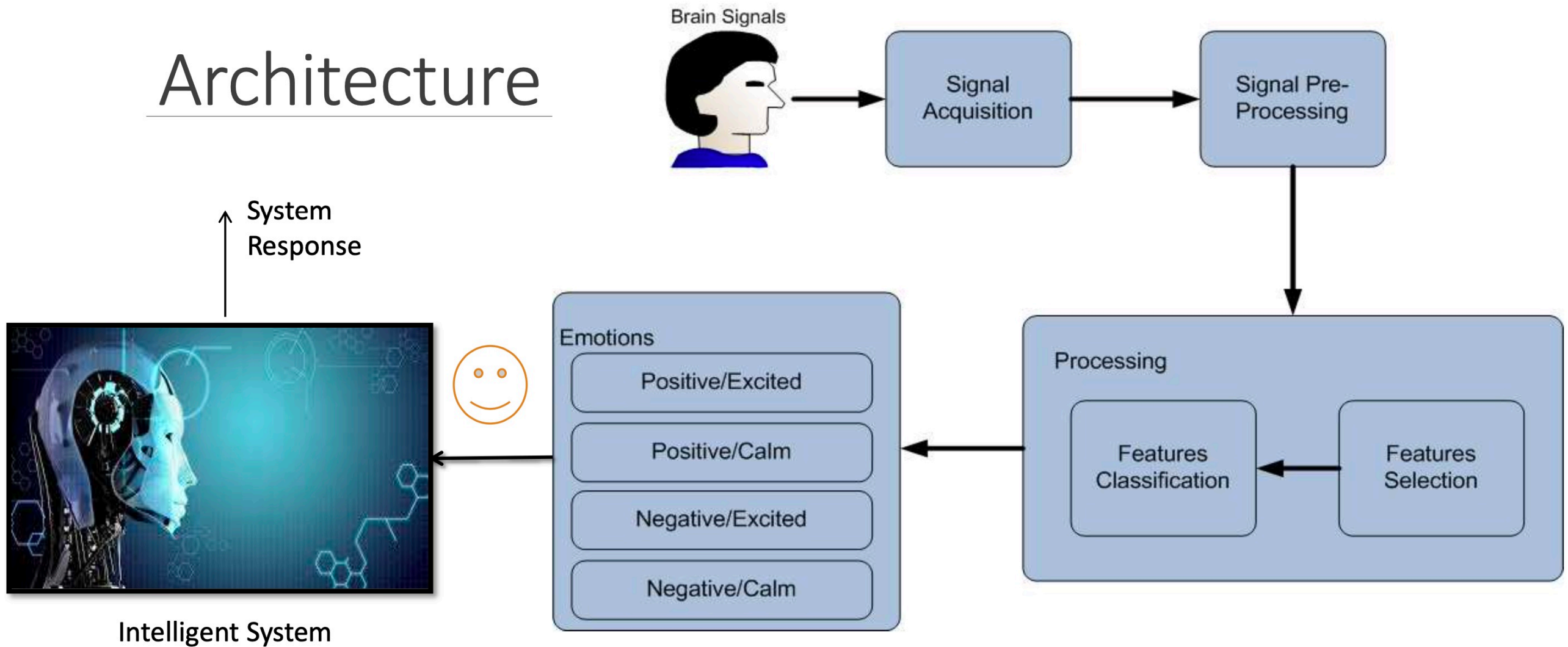
- Emotion Perception & Synthesis



Affective Computing

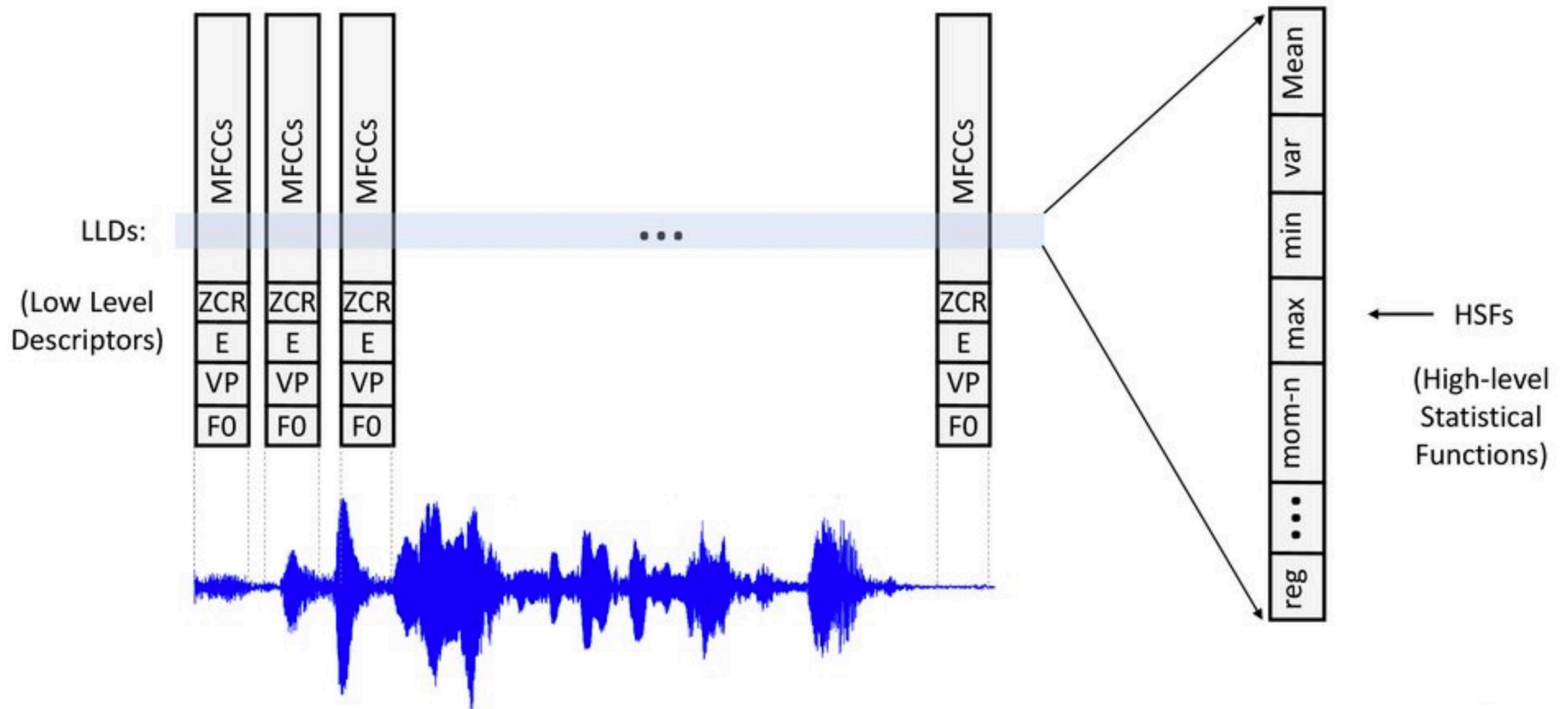
- Speech - Emotion Recognition

Architecture



Affective Computing

- Speech - Emotion Recognition



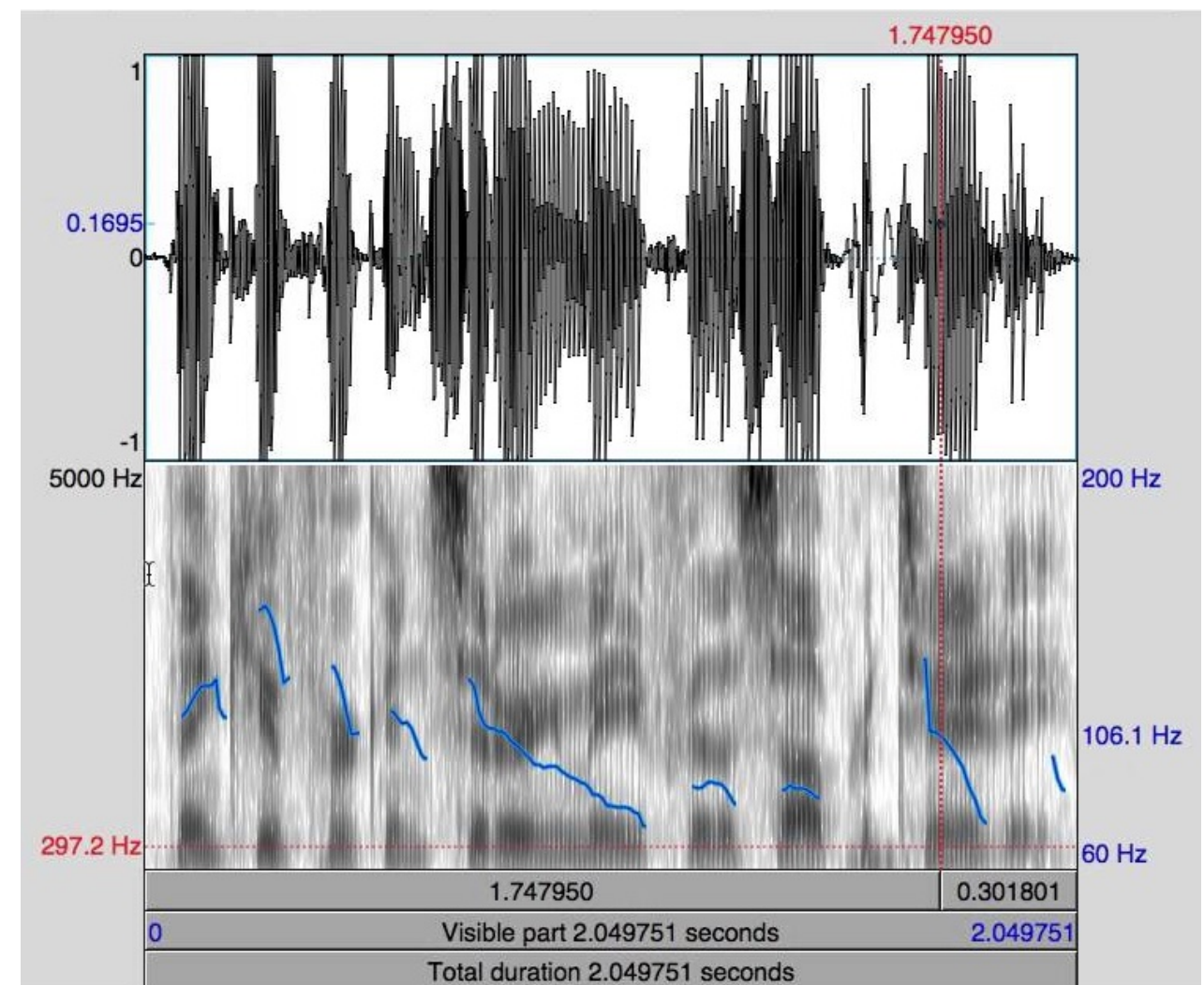
- These high-level descriptors can effectively encode emotion in speech and can be used for emotion recognition tasks.

Affective Computing

- Speech Acoustic Features

1. **Fundamental Frequency (F_0)** provides cues for the speaker's identity: **Men: 100 Hz, Women: 200 Hz, Children: 300 Hz, Crying babies: 500 Hz**. There are several algorithms for extracting the F_0 in the frequency domain, like the YAAPT algorithm (Yet Another Algorithm of Pitch Tracking).

Fundamental frequency tracking of a speech signal using Praat software



Affective Computing

- Speech Acoustic Features

2. **Mel-Frequency Cepstral Coefficients (MFCC)**: describe the **overall shape of a spectral envelope**. The cepstrum is derived by taking the log of the frequency spectrum of the audio signal. Afterward, we take the cosine transform of the list of mel-log powers, then the MFCCs are the amplitudes of the resulting spectrum.
3. **Zero Crossing Points (ZCR)**: where the waveform crosses the x-axis.
4. **Intensity**
5. **Duration**

Affective Computing

- Facial Expressions



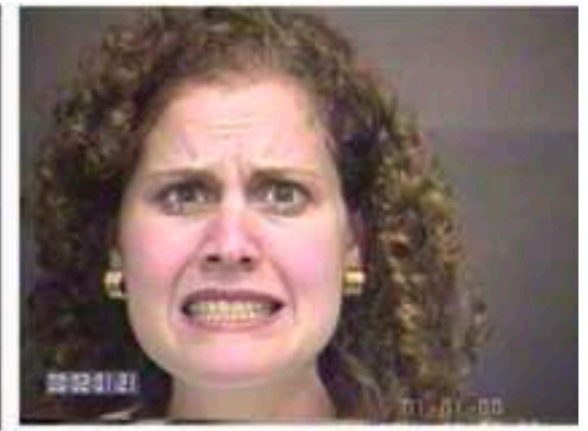
Angry



Contempt



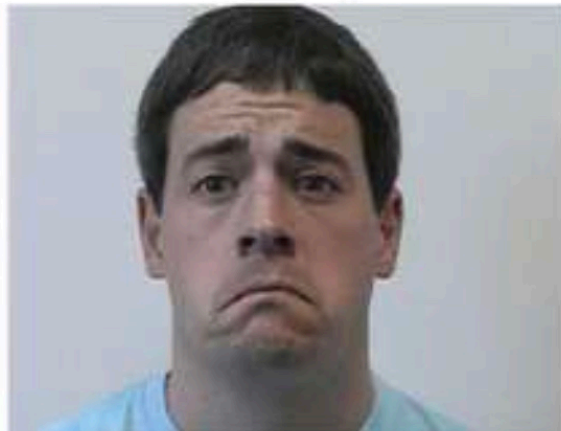
Disgust



Fear



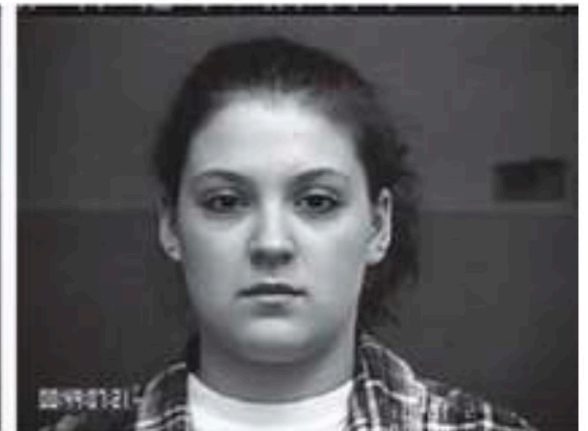
Happy



Sadness



Surprise



Neutral

Affective Computing

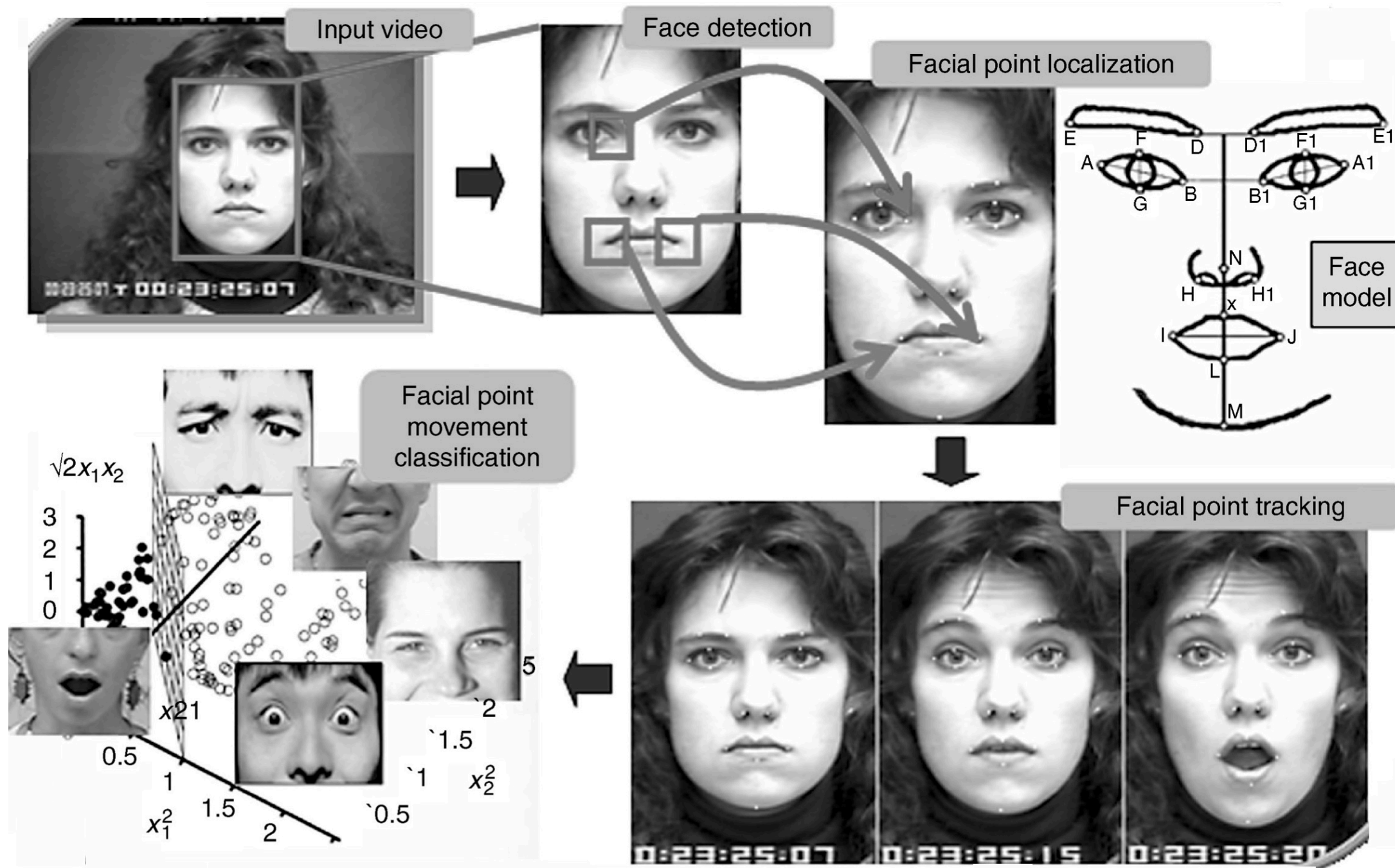
- Facial Expressions - Emotion Recognition



1. Face is extracted from the background
2. Facial geometry (e.g. locations of eyes, nose, mouth) can be estimated.
3. Based on facial geometry, facial expressions can be normalized, taking out the impact of head rotations or other head movements

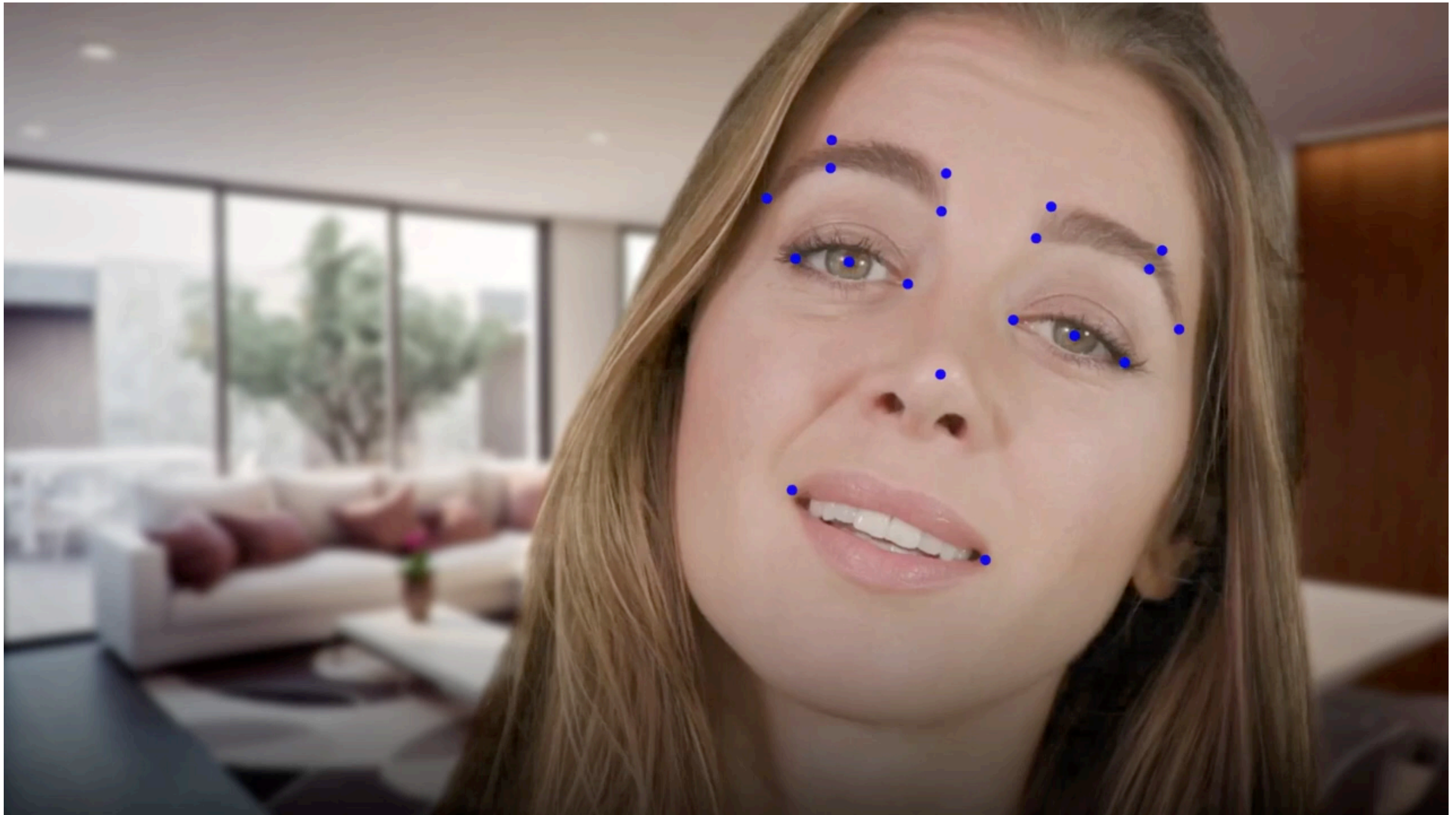
Affective Computing

- Facial Expressions - Emotion Recognition



Affective Computing

- Facial Expressions - Emotion Recognition



Emotion Tracking through a Video

Affective Computing

- Applications

1. **E-learning:** on computers wherein computing the users' feelings can be used to adjust the presentation.
2. **Marketing:** measure customer reactions to their campaigns, new products, and services to optimize their marketing strategies.
3. **Training & Tracking:** employees' stress and anxiety levels during the job and observe if they are satisfied with their current tasks and workload.
4. **Robotics:** can be used in social interaction to recognize the emotion of human users.
5. **Health Care:** can be used in counseling sessions to track and understand mental states better and help doctors support patients more effectively.
6. **Vehicle Driving:** can be used to recognize the fatigue of the driver.
7. **And more ...**

Summary

- Today we discussed

- 1- Emotional Intelligence (affective computing)

- Next week

- 1- We will discuss a different topic in social robotics; i.e., Spatial Interaction.

Readings

- Picard, Rosalind Wright. Affective Computing, 1995.

<http://affect.media.mit.edu/pdfs/95.picard.pdf>

- C. Bartneck and T. Belpaeme, et al., Human-Robot Interaction, Cambridge University Press, 2020.